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Unit 1 - Exploring the Core

PDF Documents

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UNIT 1 EXPLORING THE CORE

Topic Investigation

WHAT IS A SMART FARM?

A smart farm is like a high-tech outdoor laboratory where crops are the main experiment. Here, technology takes the lead in caring for plants. Imagine using gadgets like sensors to check how thirsty the plants are or using robots to gently tuck seeds into the soil. Computers help keep an eye on everything, ensuring the crops grow healthy and strong.

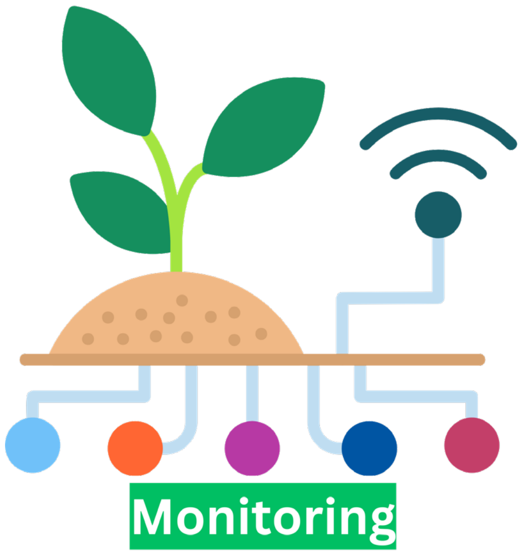
ESSENTIAL TOOLS OF SMART FARMS

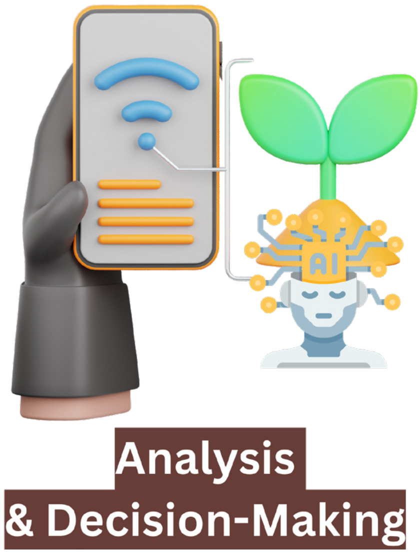
- **Eyes and Ears (Sensors):** These gadgets act as the farm's senses, constantly checking on the plants' surroundings. They monitor the warmth of the sun, the moisture in the soil, how much light is shining down, and even the nutrients available for the plants.
- **Smart Brain (Artificial Intelligence):** This is the farm's decision-maker. It takes in all the information from the sensors and figures out exactly what the plants need at any moment, like more water, light, or even when it's time to harvest.
- **Hard Workers (Robots):** These are the muscle on the farm. They carry out tasks like sowing seeds, watering rows of young plants, and keeping harmful pests away, all with precision and care.

3 MAIN PROCESSES OF SMART FARMING

Process 1) Monitoring

This is where the "Eyes and Ears (Sensors)" come into play. In this process, various sensors continuously collect data about the farm's environment, such as soil moisture, temperature, light intensity, and nutrient levels. This constant surveillance helps in understanding the exact conditions of the crops at any given time.





Process 2) Analysis and Decision-Making

The "Smart Brain (Artificial Intelligence)" takes over in this phase. It processes the data gathered by the sensors to make informed decisions about what the plants need. This could involve determining the best time to water the crops, when to apply nutrients, or identifying the optimal growth conditions for various plants.

Process 3) Action

Here, the "Hard Workers (Robots)" step in to carry out the decisions made by the artificial intelligence. Whether it's adjusting water levels, planting seeds, applying fertilizers, or protecting the crops from pests, these robots ensure that each plant receives tailored care according to the insights derived from the monitoring and analysis phases.

ADVANTAGES OF SMART FARMING

- **Bountiful Harvests:** Smart farms can turn even small patches of land into productive gardens, yielding more crops than traditional farming.
- **Healthier Food:** Because smart farms can grow food in cleaner, more controlled environments, the produce is not only tastier but also healthier.
- **Environmental Guardians:** By using resources like water and fertilizers more efficiently and cutting down on chemicals, smart farms are friends of the Earth.
- **Simplified Farming:** Imagine running a farm from a smartphone or tablet! With robots and AI, the hard work is handled by technology, making farming more accessible and enjoyable for everyone.

TIPS

Why do you think smart farms are becoming more and more important?

Inquiry & Reflection

Let's explore different environments where we could set up a smart farm and think about the types of robots that would help manage each specific condition like temperature, humidity, nutrition, noise, and light.

Environment	Robots Needed & Why
Temperature	1) A robot that can check how hot or cold it is, kind of like a walking thermometer. 2) A robot that thinks about the temperature and decides what to do next, like if the plants need cooling down. 3) A moving robot that can go around giving water to the plants when it's too hot.
Humidity	
Nutrition	
Noise	
Light	